BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MULTAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 2481 / 2482

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

CHEMISTRY PAPER-I GROUP-I (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 2481

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. The largest number of molecules are present in:

(A) 3.6g of H_2O
(B) 4.8g of C_2H_5OH
(C) 2.8g of CO
(D) 5.4g of N_2O_5

Q2. The number of isotopes of nickel are:

(A) 1
(B) 3
(C) 5
(D) 7

Q3. Iodine dissolved in water in presence of KI is due to formation of given species:

(A) I_2
(B) I'
(C) I_4^{-1}
(D) I_3^{-1}

Q4. The comparative rates at which solute moves in paper chromatography depends on:

(A) Size of paper
(B) R_f value of solute
(C) Temperature of experiment
(D) Size of chromatographic tank

Q5. Which of given will have highest rate of diffusion?

(A) O_2
(B) CO_2
(C) NH_3
(D) SO_2

Q6. Molar volume of CO_2 is maximum at:

(A) S.T.P
(B) 127°C and 1 atm
(C) 0°C and 2 atm
(D) 273°C and 2 atm

Q7. At Murree hills water boils at:

(A) 98°C
(B) 100°C
(C) 0°C
(D) 50°C

Q8. The molecules of CO_2 in dry ice form the:

(A) Ionic crystals
(B) Covalent crystals
(C) Molecular crystals
(D) Any type of crystals

Q9. Orbitals having same energy are called:

(A) Hybrid orbitals
(B) Valence orbitals
(C) d-orbitals
(D) Degenerate orbitals

Q10. The nature of positive rays depends upon:

(A) The nature of electrodes
(B) Nature of discharge tube
(C) The nature of residual gas
(D) All of these

Q11. Which of given species has unpaired electrons in anti-bonding molecular orbitals?

(A) N_2^{-2}
(B) O_2^{+2}
(C) B_2
(D) F_2

Q12. The bond order of N_2 molecule according to Molecular Orbital Theory is:

(A) Zero
(B) 01
(C) 02
(D) 03

Q13. For a given process the heat changes at constant pressure(q_p) and at constant volume(q_v) are related to each other as:

(A) q_p = q_v
(B) q_p < q_v
(C) q_p > q_v
(D) q_p = q_v/2

Q14. pH of pure water is:

(A) 7.0
(B) 5.4
(C) 4.4
(D) 8.0

Q15. Molarity of pure water is:

(A) 1
(B) 18
(C) 55.5
(D) 6

Q16. If salt bridge is not used between two half cells then the voltage:

(A) Decreases rapidly
(B) Decreases slowly
(C) Does not change
(D) Drops to zero

Q17. The unit of rate constant is same as that of rate of reaction in:

(A) First order reaction
(B) Second order reaction
(C) Zero order reaction
(D) Third order reaction

CHEMISTRY PAPER-I GROUP-I (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts.

(i) Differentiate between ion and molecular ion.

(ii) What are macromolecules? Give an example.

(iii) What is justification of two strong peaks in the mass spectrum for bromine while for iodine only one peak at 127 a.m.u is indicated?

(iv) Define sublimation. Give two examples.

(v) Give two applications of paper chromatography.

(vi) What do you mean by distribution coefficient?

(vii) What is absolute zero?

(viii) Give two conditions when gases deviate from ideal behaviour.

(ix) Joule Thomson effect produces cooling. How?

(x) How can direction of a reversible reaction be predicted by K_C?

(xi) Define Buffer capacity.

(xii) Give two applications of solubility product.

Q3. (Marks 16) Attempt any eight parts.

(i) Why boiling point of H_2O is greater than HF?

(ii) Evaporation causes cooling. Justify.

(iii) What is habit of crystal? What is the effect of impurity on shape of crystal?

(iv) Define Allotropy. Write names of two elements showing allotropy.

(v) Why positive rays are called canal rays?

(vi) How dual nature of matter was got verified by Davison and Germer experimentally?

(vii) How slow neutrons are used to carry out radioactive reactions by emitting gamma radiations?

(viii) Cathode rays travel in a straight line. Justify with diagram.

(ix) What is the effect of temperature on phenol-water system?

(x) Why $AlCl_3$ and $CuSO_4$ give acidic solution in water? Give chemical equation of each.

(xi) What is dilatometric and refractrometric method for the determination of concentration of reactant?

(xii) What is activation of catalyst?

Q4. (Marks 12) Attempt any six parts.

(i) What is dipole moment? Give its various units.

(ii) What is octet rule? Give two examples of compounds which deviate from it?

(iii) Write the Lewis structures for (i) CS_2 (ii) HCN

(iv) How do you compare the bond strengths of polar and non-polar molecules?

(v) Justify that the burning of a candle is a spontaneous process.

(vi) Define state and state functions with examples.

(vii) Differentiate between heat and temperature.

(viii) A salt bridge maintains the electrical neutrality in the cell. Why?

(ix) Calculate the oxidation number of "S" in $Cr_2(SO_4)_3$.

Q. SECTION-II

Q. (Marks 24) NOTE: Attempt any three questions.

Q5.

(a) [1+1+2=4] Define and differentiate between actual and theoretical yield. Explain why actual yield is usually less than theoretical yield.

(b) [3+1=4] Describe with the help of diagram the monometric method for the measurement of vapour pressure.

Q6.

(a) [4] 250cm³ of the sample of hydrogen effuses four times as rapidly as 250cm³ of a unknown gas. Calculate the molar mass of unknown gas.

(b) [4] State and explain main points of Planck's quantum theory.

Q7.

(a) [4] What is ionization energy? Discuss the variation of ionization energy in the periodic table.

(b) [4] The solubility product of Ag_2CrO_4 is 2.6×10^{-2} at 25°C. Calculate the solubility of the compound.

Q8.

(a) [4] Explain the glass calorimetric method for the measurement of enthalpy of reaction.

(b) [4] Define standard electrode potential. Explain the measurement of electrode potential of copper.

Q9.

(a) [4] Explain Beckmann's Method for measuring depression in freezing point.

(b) [4] Explain the effect of temperature on the rate constant of a reaction by "Arrhenius Equation".

GROUP – II

CHEMISTRY PAPER-I GROUP-II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 2482

> Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. The number of isotopes of tin(Sn) is:

(A) Six
(B) Nine
(C) Eleven
(D) Three

Q2. 27g of Al will react completely with how much mass of O_2 to produce Al_2O_3?

(A) 8g of oxygen
(B) 16g of oxygen
(C) 32g of oxygen
(D) 24g of oxygen

Q3. A complete quantitative analysis consists of how many major steps:

(A) Two
(B) Five
(C) Four
(D) Three

Q4. Solvent extraction method is particularly useful technique for separation, when the product to be separated is:

(A) Non volatile or thermally unstable
(B) Volatile or thermally stable
(C) Non volatile or thermally stable
(D) Volatile or thermally unstable

Q5. The melting point of ice on Kelvin Scale, at 1 atmospheric pressure is:

(A) 273K
(B) 373K
(C) 0K
(D) -273 K

Q6. Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of total pressure exerted by oxygen is:

(A) 1/9
(B) 8/9
(C) 1/3
(D) 16/17

Q7. In order to maintain the boiling point of water at 110°C, the external pressure should be:

(A) Between 200 torr and 760 torr
(B) Between 760 torr and 1200 torr
(C) 765 torr
(D) Any value of torr

Q8. Ionic solids are characterized by:

(A) Low melting points
(B) Good conductivity in solid state
(C) High vapour pressure
(D) Solubility in polar solvents

Q9. Splitting of spectral lines when atoms are subjected to strong electric field is called:

(A) Zeeman effect
(B) Stark effect
(C) Compton effect
(D) Photoelectric effect

Q10. Quantum number values for 2p orbitals are:

(A) $n = 2, \ell = 1$
(B) $n = 1, \ell = 2$
(C) $n = 1, \ell = 0$
(D) $n = 2, \ell = 0$

Q11. The most electronegative element of the periodic table is:

(A) Chlorine
(B) Bromine
(C) Fluorine
(D) Iodine

Q12. Which molecule shows linear molecular geometry?

(A) CH_4
(B) $AlCl_3$
(C) H_2O
(D) $BeCl_2$

Q13. Calorie is equivalent to:

(A) 0.4184 J
(B) 41.8 J
(C) 4.18 J
(D) 418.4 J

Q14. The solubility product of AgCl is 2.0×10^{-10} mole^{2}dm^{-6}. The maximum concentration of Ag^+ ions in the solution is:

(A) 2.0×10^{-10} mole dm^{-3}
(B) 1.41×10^{-5} mole dm^{-3}
(C) 1.0×10^{-10} mole dm^{-3}
(D) 4.0×10^{-20} mole dm^{-3}

Q15. A solution of glucose is 10% w/v. The volume in which 1g mole of it is dissolved will be:

(A) 1.8dm^3
(B) 1dm^3
(C) 200cm^3
(D) 900cm^3

Q16. If the salt bridge is not used between two half cells, the voltage:

(A) Decreases rapidly
(B) Decreases slowly
(C) Does not change
(D) Drops to zero

Q17. With increase of 10°C temperature, the rate of reaction doubles. This increase in rate of reaction is due to:

(A) Decrease in activation energy of reaction
(B) Decrease in the number of collisions between reactant molecules
(C) Increase in effective collisions
(D) Increase in activation energy of reactants

CHEMISTRY PAPER-I GROUP-II (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts.

(i) What is the principle of mass spectrometry?

(ii) 23g of sodium and 238g of uranium have equal number of atoms in them. Give the reason.

(iii) Define gram atom. Give two examples.

(iv) What is R_f value? Give its formula.

(v) Write down the four main characteristics of the solvent used for crystallization.

(vi) What is solvent extraction? Give its importance.

(vii) Gases deviate more significantly at high pressure and low temperature. Why?

(viii) How do you differentiate between effusion and diffusion of gases?

(ix) Derive Graham's law of diffusion from kinetic molecular theory of gases.

(x) Give two applications of common ion effect.

(xi) Why do the equilibrium constant value has its units for some of the reversible reactions but has no units for some other reactions?

(xii) How can we prepare acidic buffers? Give an example.

Q3. (Marks 16) Attempt any eight parts.

(i) Why water is liquid but hydrogen sulphide is gas at room temperature?

(ii) How dynamic equilibrium is established in a close vessel?

(iii) Why molar heat of vapourization is greater than molar heat of fusion?

(iv) Why liquid crystals are used as temperature sensors?

(v) Write reason for production of positive rays.

(vi) How will you justify that cathode rays move in straight line?

(vii) Write any two postulates of Plank's Quantum Theory.

(viii) Calculate wave number of second spectral line of Balmer series.

(ix) NaCl and KNO_3 are used to lower the melting point of ice. Why?

(x) Why non-ideal solutions do not obey Raoult's law?

(xi) Differentiate between rate of reaction and rate constant of a reaction.

(xii) What is heterogenous catalysis? Give an example.

Q4. (Marks 12) Attempt any six parts.

(i) Impure copper can be purified by electrolytic process. Explain by giving reason.

(ii) Calculate the oxidation number of underlined elements $Cr_2(SO_4)_3$; Na_2CO_3.

(iii) What is Thermochemistry? Give examples.

(iv) Define enthalpy of reaction. Give example.

(v) In case of liquids and solids system $\Delta H \approx \Delta E$. Explain.

(vi) Bond distance is the compromise distance between two atoms. Explain with reason.

(vii) The melting points of electrovalent compounds are higher than covalent compounds. Explain with reason.

(viii) There is no bond in chemistry with 100% ionic character. Explain.

(ix) The atomic radius cannot be measured precisely. Explain with reason.

Q. SECTION-II

Q. (Marks 24) NOTE: Attempt any three questions.

Q5.

(a) [4] Define stoichiometry. Write two assumptions of stoichiometry. Give example.

(b) [4] Define boiling point. Explain effect of external pressure on boiling point.

Q6.

(a) [4] Calculate the mass of 1 dm^3 of NH_3 gas at 30°C and 1000 mm Hg pressure considering that NH_3 behaving ideally.

(b) [4] Differentiate between atomic emission and atomic absorption spectrum with diagram.

Q7.

(a) [4] What is paramagnetic character? Give the reason for paramagnetic character of oxygen (O_2) through orbital picture.

(b) [4] The solubility of CaF_2 in water at 25°C is found to be 2.05×10^{-4} mol dm^{-3}. What is the value of K_SP at this temperature?

Q8.

(a) [3+1=4] Explain construction and working of Bomb calorimeter. Write also formula to calculate q.

(b) [1+3=4] Define electrochemical series. Write its any two applications.

Q9.

(a) [4] What do you mean by elevation of boiling point? Explain Landsberger's method for its measurement.

(b) [2+2=4] Discuss Half Life Method and method of large excess to find order of a reaction.